A Multiterminal Information Bottleneck for Distributed Bird's-Eye-View Perception

Rida Cao, Ruihao Wang, Huihui Wu, Tao Guo, Xiaoyu Zhao and Qiaoqiao Zhou

***Abstract*—Existing bird's-eye-view (BEV) detectors typically assume centralized access to sensor features, whereas communication-constrained perception rarely accounts for heterogeneous modality roles and cross-modal redundancy. We study distributed camera–LiDAR–radar perception, where each sensor encodes BEV features for two overlapping decoders: a camera–LiDAR semantic–geometric decoder and a LiDAR–radar motion decoder sharing one LiDAR description. We formulate this setting as a multiterminal rate–distortion–relevance–redundancy (RDRR) problem and derive inner and outer bounds for discrete memoryless sources. Based on this formulation, we propose RDRR-BEV, a redundancy-aware distributed framework that realizes a multiterminal information bottleneck using compact stochastic BEV representations to preserve task-relevant information and suppress duplicated content. Its objective combines detection loss, latent complexity, task relevance, and a total-correlation penalty, with information terms approximated by differentiable soft-entropy surrogates. On the nuScenes mini dataset, RDRR-BEV reduces the soft-entropy rate proxy by 13.7% relative to a rate-only baseline while improving detection accuracy and velocity estimation, and maintains more accurate velocity estimation under severe channel noise without SNR-specific retraining.**

I. Introduction

Bird's-eye-view (BEV) perception [1], [2] maps heterogeneous sensor observations into a common spatial representation for multi-sensor fusion, supporting object classification, localization, and motion estimation. Specifically, cameras capture rich visual information, LiDAR supplies accurate geometry, and radar measures radial velocity. Representative methods fuse camera with LiDAR or radar [3], [4], with BEVFusion [5] aligning heterogeneous features in a shared BEV space. However, these methods generally assume that all modality features are locally available at the fusion receiver. This assumption need not hold in distributed or edge-assisted perception architectures, where modality-specific representations must be communicated before fusion [6]. Furthermore, ego-vehicle multi-modal perception exploits the complementary information provided by onboard cameras, LiDAR, and radar, which can help mitigate occlusions and motivate the rate-constrained transmission of task-relevant features.

Multiterminal source coding studies the distributed compression of correlated observations [7], [8]. Distributed information bottleneck and task-oriented communication further seek compact descriptions that preserve task-relevant information under rate or channel constraints [9], [10], [11]. However, existing task-oriented formulations generally do not simultaneously account for heterogeneous modality roles, task-dependent decoding requirements, and cross-modal redundancy arising when multiple sensors observe the same scene. Such uncontrolled redundancy leads to repeated transmission of shared scene content and thus wastes communication resources. Imposing a separate rate constraint on each description does not address this problem, since it neither controls the dependence among descriptions nor reflects the task-dependent utility of each one.

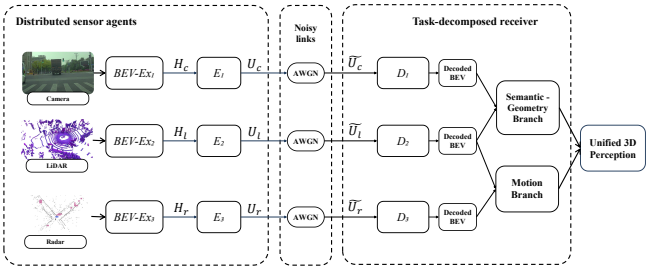


Fig. 1. Overview of the proposed RDRR-BEV framework. The camera–LiDAR branch predicts semantic and geometric attributes, whereas the LiDAR–radar branch estimates object velocity. The decoded LiDAR representation is shared by both branches.

Motivated by these limitations, we consider distributed camera–LiDAR–radar perception with three modality-specific agents on the same sensing platform that separately encode their observations without inter-agent communication as shown in Fig. 1. The semantic–geometric decoder uses the camera and LiDAR descriptions, whereas the motion decoder uses the LiDAR and radar descriptions. Accordingly, the two decoder sets overlap in the LiDAR description, which is transmitted once and reused by both branches, thereby coupling their rate requirements and motivating a redundancy-controlled multiterminal information bottleneck that preserves branch-relevant information while limiting duplicated content across the three descriptions.

In this paper, we formulate the resulting distributed BEV setting as a multiterminal rate–distortion–relevance–redundancy (RDRR) problem with overlapping decoder sets and a shared description. Particularly, for the corresponding discrete memoryless source model, we derive an achievable inner bound and a converse outer bound on the RDRR region. Guided by this formulation, we propose RDRR-BEV, a

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task-decomposed and redundancy-aware distributed perception framework that instantiates a multiterminal information bottleneck over noisy communication links. Its training objective jointly accounts for task distortion, a description-complexity surrogate for rate, task relevance, and cross-modal redundancy. We measure cross-modal redundancy by the total correlation [12] among the learned camera, LiDAR, and radar descriptions, which captures their global dependence in a single quantity. Since mutual information and total correlation are difficult to estimate directly for high-dimensional representations, we use soft entropy [13] to construct differentiable proxies for description complexity, task relevance, and cross-modal redundancy.

The main contributions are summarized as follows:

- We formulate the distributed camera–LiDAR–radar BEV perception as an RDRR problem. For the corresponding discrete memoryless source model, we derive an achievable inner bound and a converse outer bound on the RDRR region.
- We propose RDRR-BEV, a distributed camera–LiDAR–radar BEV perception framework that combines a task-decomposed receiver with a redundancy-controlled multiterminal information-bottleneck objective.
- Experiments on the nuScenes v1.0 mini dataset [14] validate the complementary benefits of redundancy control and task decomposition, and show that RDRR-BEV remains robust across noise levels without SNR-specific retraining.

II. System Model

As shown in Fig. 1, let $\mathcal{A} = \{c, l, r\}$ index the camera, LiDAR, and radar agents, respectively. Agent $i \in \mathcal{A}$ observes only its local BEV source H_i, while Y_{sg} and Y_{mot} denote the semantic–geometric and motion targets, respectively. Throughout, uppercase letters denote random variables, while lowercase letters denote their realizations. Specifically, h_i denotes a realization of H_i for $i \in \mathcal{A}$, whereas y_{sg} and y_{mot} denote realizations of Y_{sg} and Y_{mot}, respectively. Together with the semantic–geometric target Y_{sg} and the motion target Y_{mot}, the sources follow an arbitrary joint distribution $p(h_c, h_l, h_r, y_{\mathrm{sg}}, y_{\mathrm{mot}})$, where no independence is assumed among the modalities.

For a length-n discrete-memoryless source, agent i independently maps its observation to a finite-alphabet message

$$M_i = e_i^{(n)}(H_i^n) \in \mathcal{M}_i, \qquad R_i^{(n)} = \frac{1}{n}\log_2 |\mathcal{M}_i|, \quad (1)$$

where M_i denotes the length-n discrete message used for operational-rate and block-coding constraints. Note that in Fig. 1, U_i denotes the corresponding single-letter task-oriented description used in the information-bottleneck formulation of Section III. Therefore, M_i supports coding analysis, while U_i represents the learned description.

For the theoretical analysis, each physical link is represented by its induced description rate R_i, while noisy finite-dimensional transmission is evaluated in Section V.

The heterogeneous sensing roles induce two overlapping decoder sets. The semantic–geometric decoder uses the camera and LiDAR messages, whereas the motion decoder uses the LiDAR and radar messages:

$$\begin{aligned} \widehat{Y}_{\mathrm{sg}}^n &= \psi_{\mathrm{sg}}^{(n)}(M_c, M_l), \\ \widehat{Y}_{\mathrm{mot}}^n &= \psi_{\mathrm{mot}}^{(n)}(M_l, M_r). \end{aligned} \quad (2)$$

Thus, the LiDAR message is generated and communicated only once but must simultaneously support both tasks.

Since the two decoder sets overlap at the LiDAR message, we define the branch-wise message redundancy over the length-n block by

$$I(M_l; M_c) + I(M_l; M_r). \quad (3)$$

This quantity measures the aggregate statistical dependence between the shared LiDAR message and the two branch-specific messages over the entire block. In the operational constraints below, it is normalized by n to obtain a per-source-symbol redundancy measure. Its single-letter counterpart and its learnable global surrogate are developed in Section III.

Let d_{sg} and d_{mot} denote the single-letter distortion measures, with corresponding budgets D_{sg} and D_{mot}. The relevance requirements are η_{sg} and η_{mot}, and Γ is the maximum per-source-symbol branch-wise message-redundancy budget. A tuple

$$(R_c, R_l, R_r, D_{\mathrm{sg}}, D_{\mathrm{mot}}, \eta_{\mathrm{sg}}, \eta_{\mathrm{mot}}, \Gamma)$$

is achievable if, for every $\epsilon > 0$ and sufficiently large n, there exists a distributed code satisfying

$$\begin{aligned} \frac{1}{n}\mathbb{E}\left[\sum_{t=1}^{n} d_{\mathrm{sg}}\big(Y_{\mathrm{sg},t}, \widehat{Y}_{\mathrm{sg},t}\big)\right] &\le D_{\mathrm{sg}} + \epsilon, \\ \frac{1}{n}\mathbb{E}\left[\sum_{t=1}^{n} d_{\mathrm{mot}}\big(Y_{\mathrm{mot},t}, \widehat{Y}_{\mathrm{mot},t}\big)\right] &\le D_{\mathrm{mot}} + \epsilon, \\ \frac{1}{n} I(Y_{\mathrm{sg}}^n; M_c, M_l) &\ge \eta_{\mathrm{sg}} - \epsilon, \\ \frac{1}{n} I(Y_{\mathrm{mot}}^n; M_l, M_r) &\ge \eta_{\mathrm{mot}} - \epsilon, \\ \frac{1}{n}\left[I(M_l; M_c) + I(M_l; M_r)\right] &\le \Gamma + \epsilon, \\ \frac{1}{n}\log_2 |\mathcal{M}_i| &\le R_i + \epsilon, \qquad i \in \mathcal{A}. \end{aligned} \quad (4)$$

The first two inequalities constrain task distortion, the next two enforce task relevance for the decoder branches, the fifth limits normalized branch-wise redundancy, and the last specifies the individual description rates.

For fixed $(D_{\mathrm{sg}}, D_{\mathrm{mot}}, \eta_{\mathrm{sg}}, \eta_{\mathrm{mot}}, \Gamma)$, the closure of all achievable rate triples defines the rate–distortion–relevance–redundancy region studied in Section IV.

III. RDRR-BEV Framework

A. Task-Decomposed BEV Perception

For the information-bottleneck formulation, let U_i denote the single-letter task-oriented description generated by agent $i \in \mathcal{A} = \{c, l, r\}$, as shown in Fig. 1. Since the three agents

independently encode their local observations without inter-agent communication, their descriptions satisfy

$$p(u_c, u_l, u_r \mid h_c, h_l, h_r) = \prod_{i \in \mathcal{A}} p(u_i \mid h_i). \tag{5}$$

Following the decoder sets in Section II, the two task reconstructions are

$$\widehat{Y}_{\mathrm{sg}} = \varphi_{\mathrm{sg}}(U_c, U_l), \qquad \widehat{Y}_{\mathrm{mot}} = \varphi_{\mathrm{mot}}(U_l, U_r). \tag{6}$$

The receiver uses two task-specific branches: the semantic–geometric branch combines U_c and U_l to predict object categories, locations, and geometric attributes, whereas the motion branch combines U_l and U_r to estimate velocity. The LiDAR description U_l is transmitted once and shared by both branches.

B. Multiterminal Information Bottleneck with Redundancy Control

To jointly optimize task fidelity and communication efficiency under distributed encoding, we formulate a constrained multiterminal information bottleneck. It limits the aggregate description complexity, enforces task-specific relevance for both decoder branches, and bounds the branch-wise redundancy induced by the shared LiDAR description. Let $C_0 \geq 0$ denote the prescribed upper bound on description complexity. The resulting constrained information bottleneck problem is

$$\begin{aligned} \min \quad & \mathbb{E}[d_{\mathrm{sg}}(Y_{\mathrm{sg}}, \widehat{Y}_{\mathrm{sg}})] + \mathbb{E}[d_{\mathrm{mot}}(Y_{\mathrm{mot}}, \widehat{Y}_{\mathrm{mot}})] \\ \text{s.t.} \quad & \sum_{i \in \mathcal{A}} I(H_i; U_i) \leq C_0, \\ & I(Y_{\mathrm{sg}}; U_c, U_l) \geq \eta_{\mathrm{sg}}, \qquad I(Y_{\mathrm{mot}}; U_l, U_r) \geq \eta_{\mathrm{mot}}, \\ & I(U_l; U_c) + I(U_l; U_r) \leq \Gamma. \end{aligned} \tag{7}$$

However, the branch-wise redundancy constraint only measures the dependence between the shared LiDAR description and the two branch-specific descriptions. To capture the global dependence among all three modality descriptions, we further use total correlation:

$$\begin{aligned} \mathcal{T}(U_c, U_l, U_r) &= D_{\mathrm{KL}}(p(u_c, u_l, u_r) \| p(u_c)p(u_l)p(u_r)) \\ &= H(U_c) + H(U_l) + H(U_r) - H(U_c, U_l, U_r) \\ &= I(U_l; U_c) + I(U_l; U_r) + I(U_c; U_r \mid U_l). \end{aligned} \tag{8}$$

The last equality shows that total correlation includes the two branch-wise dependencies involving the shared LiDAR description and additionally captures the residual camera–radar dependence conditioned on LiDAR. Let β, γ_{sg}, γ_{mot}, and λ be nonnegative coefficients controlling description complexity, task relevance, and global redundancy, respectively. For convenience, define the task-distortion loss as

$$\mathcal{L} \triangleq \mathbb{E}[d_{\mathrm{sg}}(Y_{\mathrm{sg}}, \widehat{Y}_{\mathrm{sg}})] + \mathbb{E}[d_{\mathrm{mot}}(Y_{\mathrm{mot}}, \widehat{Y}_{\mathrm{mot}})]. \tag{9}$$

The resulting ideal redundancy-aware multiterminal information-bottleneck objective is

$$\begin{aligned} \min \quad & \mathcal{L} + \beta \sum_{i \in \mathcal{A}} I(H_i; U_i) - \gamma_{\mathrm{sg}} I(Y_{\mathrm{sg}}; U_c, U_l) \\ & - \gamma_{\mathrm{mot}} I(Y_{\mathrm{mot}}; U_l, U_r) + \lambda \mathcal{T}(U_c, U_l, U_r). \end{aligned} \tag{10}$$

This objective jointly minimizes task distortion and description complexity, preserves information relevant to the two task branches, and penalizes global dependence among the three transmitted descriptions.

C. Soft-Entropy-Based Realization

The information quantities in (10) are difficult to estimate directly for high-dimensional continuous BEV representations. We therefore use differentiable soft entropy [13]. Let A_i denote the prototype-assignment representation of agent $i \in \mathcal{A}$, and let $\widehat{H}_{\mathrm{s}}(\cdot)$ denote the corresponding soft-entropy estimate. With Y_{bev} denoting the dense BEV supervision, we define

$$\begin{aligned} \mathcal{R}_{\mathrm{comp}} &= \sum_{i \in \mathcal{A}} \widehat{H}_{\mathrm{s}}(A_i), \\ \mathcal{R}_{\mathrm{rel}} &= \sum_{i \in \mathcal{A}} \left[\widehat{H}_{\mathrm{s}}(A_i) - \widehat{H}_{\mathrm{s}}(A_i \mid Y_{\mathrm{bev}}) \right], \\ \mathcal{R}_{\mathrm{red}} &= \sum_{i \in \mathcal{A}} \widehat{H}_{\mathrm{s}}(A_i) - \widehat{H}_{\mathrm{s}}(A_c, A_l, A_r). \end{aligned} \tag{11}$$

These terms provide surrogates for description complexity, task relevance, and global redundancy, respectively. The practical redundancy-aware training objective is

$$\mathcal{L}_{\mathrm{RDRR}} = \mathcal{L} + \beta \mathcal{R}_{\mathrm{comp}} - \gamma \mathcal{R}_{\mathrm{rel}} + \lambda \mathcal{R}_{\mathrm{red}}. \tag{12}$$

where β, γ, and λ control the three regularization terms.

The preceding formulation defines the learnable RDRR-BEV framework and its training objective. Section IV will then characterize the corresponding multiterminal rate region from an information-theoretic perspective.

IV. Achievable Inner and Outer Bounds

We now characterize the rate region induced by the system in Fig. 1. For this analysis, the aligned features and task variables follow an arbitrary joint distribution $p(h_c, h_l, h_r, y_{\mathrm{sg}}, y_{\mathrm{mot}})$; no independence is imposed across modalities. Its length-n discrete-memoryless extension is considered. Agent i maps H_i^n to a discrete message M_i at operational rate $R_i^{(n)}$. Here, M_i is the block-coding counterpart of the neural description U_i introduced in Section III.

Let Q be a finite-valued time-sharing variable and (U_c, U_l, U_r) be finite-valued auxiliary descriptions generated according to

$$\begin{aligned} & p(q, h_c, h_l, h_r, y_{\mathrm{sg}}, y_{\mathrm{mot}}, u_c, u_l, u_r) \\ & \quad = p(q)p(h_c, h_l, h_r, y_{\mathrm{sg}}, y_{\mathrm{mot}}) \prod_{i \in \mathcal{A}} p(u_i \mid h_i, q). \end{aligned} \tag{13}$$

The two reconstructions preserve the receiver structure: $\widehat{Y}_{\mathrm{sg}} = \varphi_{\mathrm{sg}}(U_l, U_c, Q)$ and $\widehat{Y}_{\mathrm{mot}} = \varphi_{\mathrm{mot}}(U_l, U_r, Q)$. Here, d_{sg} and d_{mot} denote the single-letter distortion measures for the semantic–geometric and motion tasks, respectively. An admissible choice obeys

$$
\begin{aligned}
\mathbb{E}[d_{\mathrm{sg}}(Y_{\mathrm{sg}}, \widehat{Y}_{\mathrm{sg}})] &\le D_{\mathrm{sg}}, \\
\mathbb{E}[d_{\mathrm{mot}}(Y_{\mathrm{mot}}, \widehat{Y}_{\mathrm{mot}})] &\le D_{\mathrm{mot}}, \\
I(Y_{\mathrm{sg}}; U_l, U_c \mid Q) &\ge \eta_{\mathrm{sg}}, \\
I(Y_{\mathrm{mot}}; U_l, U_r \mid Q) &\ge \eta_{\mathrm{mot}}, \\
I(U_l; U_c \mid Q) + I(U_l; U_r \mid Q) &\le \Gamma,
\end{aligned}
\tag{14}
$$

where $(D_{\mathrm{sg}}, D_{\mathrm{mot}})$ are the task-distortion budgets, $(\eta_{\mathrm{sg}}, \eta_{\mathrm{mot}})$ are the task-relevance requirements, and Γ bounds the two branch-wise pairwise redundancies.

Theorem 1 (Achievable inner bound). For every admissible distribution and pair of reconstruction functions, the following rate region is achievable:

$$
\begin{aligned}
R_c &\ge I(H_c; U_c \mid U_l, Q), \\
R_r &\ge I(H_r; U_r \mid U_l, Q), \\
R_l &\ge \max\big\{ I(H_l; U_l \mid U_c, Q), \\
&\qquad\qquad I(H_l; U_l \mid U_r, Q) \big\}, \\
R_l + R_c &\ge I(H_l, H_c; U_l, U_c \mid Q), \\
R_l + R_r &\ge I(H_l, H_r; U_l, U_r \mid Q).
\end{aligned}
\tag{15}
$$

For fixed $(D_{\mathrm{sg}}, D_{\mathrm{mot}}, \eta_{\mathrm{sg}}, \eta_{\mathrm{mot}}, \Gamma)$, let $\mathcal{R}^{\mathrm{in}}_{\mathrm{RDRR}}$ denote the closure of the union, over all admissible distributions and reconstruction functions satisfying (14), of the rate triples (R_c, R_l, R_r) satisfying (15). By Theorem 1, every rate triple in $\mathcal{R}^{\mathrm{in}}_{\mathrm{RDRR}}$ is achievable. The shared LiDAR rate R_l is counted only once, consistently with the single LiDAR message in Fig. 1.

Theorem 2 (Outer bound). Let $\mathbf{R} = (R_c, R_l, R_r)$, $\mathbf{D} = (D_{\mathrm{sg}}, D_{\mathrm{mot}})$, and $\boldsymbol{\eta} = (\eta_{\mathrm{sg}}, \eta_{\mathrm{mot}})$. Define $\mathscr{O}$ as the set of non-negative tuples $(\mathbf{R}, \mathbf{D}, \boldsymbol{\eta})$ admitting finite-valued auxiliaries $(U_l^{(\mathrm{sg})}, U_c, U_l^{(\mathrm{mot})}, U_r)$ jointly distributed with the prescribed source–task tuple, and deterministic reconstructions satisfying the following conditions under the same joint distribution.

For each branch $(b, j) \in \{(\mathrm{sg}, c), (\mathrm{mot}, r)\}$, let $\widehat{Y}_b = \varphi_b^{\mathrm{out}}(U_l^{(b)}, U_j)$ with deterministic φ_b^{out}, and require

$$
\begin{aligned}
U_l^{(b)} &- H_l - (H_j, Y_b), \\
U_j &- H_j - (H_l, Y_b), \\
(U_l^{(b)}, U_j) &- (H_l, H_j) - Y_b,
\end{aligned}
\tag{16}
$$

$$
\mathbb{E}[d_b(Y_b, \widehat{Y}_b)] \le D_b,
\tag{17}
$$

$$
I(Y_b; U_l^{(b)}, U_j) \ge \eta_b.
$$

The rate constraints are

$$
\begin{aligned}
R_c &\ge I(H_l, H_c; U_c \mid U_l^{(\mathrm{sg})}), \\
R_r &\ge I(H_l, H_r; U_r \mid U_l^{(\mathrm{mot})}), \\
R_l &\ge \max\Big\{ I(H_l, H_c; U_l^{(\mathrm{sg})} \mid U_c), \\
&\qquad\qquad I(H_l, H_r; U_l^{(\mathrm{mot})} \mid U_r) \Big\}, \\
R_l + R_c &\ge I(H_l, H_c; U_l^{(\mathrm{sg})}, U_c), \\
R_l + R_r &\ge I(H_l, H_r; U_l^{(\mathrm{mot})}, U_r).
\end{aligned}
\tag{18}
$$

The auxiliary alphabet sizes are unrestricted but finite. In particular, the auxiliary variables are not required to satisfy the product test-channel factorization used in Theorem 1.

Define

$$
\mathcal{R}^{\mathrm{out}}_{\mathrm{RDRR}}(\mathbf{D}, \boldsymbol{\eta}) \triangleq \big\{ \mathbf{R} : (\mathbf{R}, \mathbf{D}, \boldsymbol{\eta}) \in \overline{\mathscr{O}} \big\},
\tag{19}
$$

where the closure is taken in the full seven-dimensional performance space. Together with Theorem 1, for every $\Gamma \ge 0$,

$$
\mathcal{R}^{\mathrm{in}}_{\mathrm{RDRR}} \subseteq \mathcal{R}^{*}_{\mathrm{RDRR}} \subseteq \mathcal{R}^{\mathrm{out}}_{\mathrm{RDRR}}.
\tag{20}
$$

The outer bound permits two branch-specific LiDAR auxiliaries, whereas the actual system transmits a single LiDAR message. These bounds concern the block source-coding model; finite-dimensional additive white Gaussian noise (AWGN) transmission is evaluated separately in the SNR experiments of Section V.

V. Experiments And Discussions

Following the aforementioned theoretical formulation, we evaluate whether the proposed training objective improves the practical rate–performance tradeoff and the robustness to channel noise.

A. Experimental Setup

a) Dataset and Metrics: We evaluate RDRR-BEV on the nuScenes v1.0 mini dataset [14]. All compared configurations use the same data processing, detector backbone, optimizer, training schedule, and evaluation code; only the rate–redundancy objective or the task-routing structure is changed. Detection performance is evaluated using the official nuScenes mean average precision (mAP), nuScenes detection score (NDS), and mean average velocity error (mAVE).

b) Implementation and Training: The models are trained for 120 epochs with a batch size of 12. To ensure controlled comparisons, all main results are obtained with the same fixed random seed of 42. Unless otherwise specified, the main comparison uses $\beta = 10^{-4}$, $\gamma = 10^{-4}$, and $\lambda = 5 \times 10^{-6}$.

c) Soft-entropy Rate Proxy: All information quantities in (11) use the prototype-based soft entropy on natural logarithms and are thus measured in nats. The Rate column of Table I is the scaled complexity surrogate

$$
\mathrm{Rate} \triangleq \kappa R_{\mathrm{comp}} = \kappa \sum_{i \in \mathcal{A}} \widehat{H}_s(A_i), \qquad \kappa = d_z = 32,
\tag{21}
$$

Here, $d_z = 32$ is the transmitted latent-channel dimension, and we set $\kappa = d_z$ to place the per-vector soft entropy on a channel-aggregated numerical scale. This fixed calibration can be absorbed into β and does not convert the rate proxy into transmitted bits.

d) Channel Evaluation: For the channel-robustness experiment, we fix the model and vary only the signal-to-noise ratio of the additive white Gaussian noise channel over the SNR range from -5 to 25 dB. No model is retrained or adapted for an individual SNR. This setting isolates the sensitivity of the learned representations to channel perturbations.

TABLE I
MAIN RESULTS ON NUSCENES MINI UNDER THE SAME FIXED-SEED SETTING. RATE IS THE SOFT-ENTROPY PROXY.

Method	Rate↓	mAP↑	NDS↑	mAVE↓
Rate-only	4.6130	0.6246	0.5910	0.4938
Shared RDRR	4.9480	0.6212	0.6049	0.4480
RDRR-BEV	**3.9816**	**0.6383**	**0.6108**	**0.4408**
Δ_{RO}	-13.7%	$+2.2\%$	$+3.4\%$	-10.7%
Δ_{SR}	-19.5%	$+2.8\%$	$+1.0\%$	-1.6%

B. Experimental Results

1) Improved Rate–Performance Tradeoff: Table I reports the results obtained on the nuScenes mini dataset under the same fixed configuration and random seed. TransFusion [4] and BEVFusion [5] are centralized multimodal detectors that assume direct access to sensor features and do not model distributed encoders or communication-rate constraints. Their reported results are therefore not directly comparable to Table I, which presents controlled comparisons among distributed communication schemes under the same architecture and experimental protocol. To conduct fair comparisons, we take the following two different setups. Rate-only retains the task-decomposed receiver but removes the redundancy term. Shared RDRR uses the same rate–redundancy objective as RDRR-BEV but replaces task-dependent routing with a shared representation. Therefore, comparisons with Rate-only and Shared RDRR isolate the effects of redundancy regularization and task decomposition, respectively.

In Table I, Δ_{RO} and Δ_{SR} denote the relative changes of RDRR-BEV with respect to Rate-only and Shared RDRR, respectively.

Compared with Rate-only, RDRR-BEV reduces the rate proxy by approximately 13.7% and improves all three perception metrics. With the task-decomposed receiver unchanged, these gains indicate that redundancy regularization removes communication-inefficient overlap while preserving task-relevant information.

Compared with Shared RDRR, RDRR-BEV lowers the rate proxy by approximately 19.5% and improves perception, suggesting that task-specific modality routing better serves heterogeneous objectives than a shared representation.

2) Robustness to Channel Noise: We further evaluate the transmitted representations over an AWGN channel. For this experiment we use the same fixed model as in Table I, and vary only the channel SNR.

Fig. 2 evaluates the fixed, clean-trained model over AWGN channels without SNR-specific retraining or adaptation. First, the checkpoint-specific Rate proxy remains constant across SNR because it measures encoder-side description complexity rather than channel capacity. RDRR-BEV achieves a 13.7% lower rate proxy than Rate-only case (3.9816 versus 4.6130), demonstrating a consistent complexity advantage.

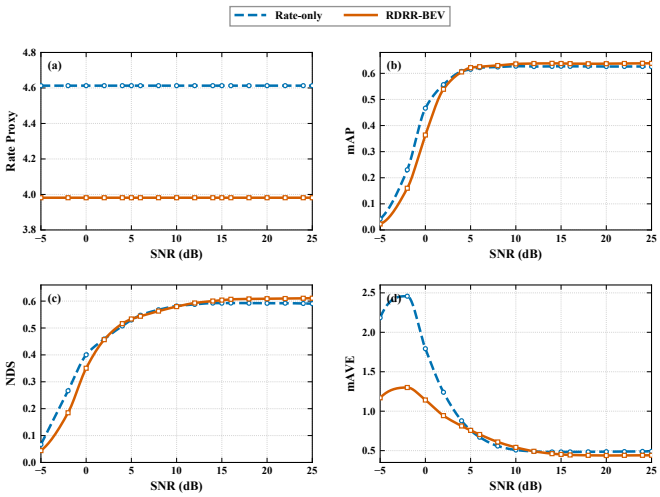


Fig. 2. Rate proxy, mAP, NDS, and mAVE of the fixed Rate-only and RDRR-BEV checkpoints versus AWGN channel SNR, without SNR-specific retraining. Higher mAP and NDS and lower rate proxy and mAVE are better.

Fig. 2 suggests three SNR regimes. At high SNRs (approximately $14 \sim 25$ dB), RDRR-BEV achieves higher mAP and NDS and lower mAVE than Rate-only while maintaining a 13.7% lower rate proxy. In the intermediate range (approximately $4 \sim 12$ dB), the metrics exhibit different crossover points: mAP favors RDRR-BEV from 5 dB, NDS shows alternating advantages near the transition, whereas mAVE favors Rate-only from 6 to 12 dB. At low SNRs ($-5 \sim 2$ dB), Rate-only achieves slightly higher mAP and NDS, plausibly because the redundancy removed by RDRR-BEV may provide incidental protection against noise. Nevertheless, RDRR-BEV maintains a substantially lower mAVE, suggesting that its LiDAR–radar motion branch preserves velocity-related cues under severe noise.

VI. CONCLUSION

This paper introduced RDRR-BEV, a distributed camera–LiDAR–radar framework for communication-constrained BEV perception. We formulated the underlying multiterminal problem with two overlapping task decoders and a shared LiDAR description, derived an achievable inner bound and a converse outer bound for the corresponding discrete memoryless model, and further developed a task-decomposed receiver trained with a redundancy-aware information-bottleneck objective based on differentiable soft-entropy proxies. On the nuScenes mini dataset, RDRR-BEV achieves a better soft-entropy rate–perception trade-off than Rate-only and Shared RDRR. Without SNR-specific retraining, it outperforms Rate-only at moderate and high SNRs and maintains more accurate velocity estimation under severe channel noise. Operating on intermediate latent representations rather than a specific backbone, the objective may generalize to other perception models and datasets.

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Appendix A
Proof of the Achievable Inner Bound

Proof. Fix a distribution and reconstruction functions satisfying the admissibility conditions of Theorem 1. Let

$$(b, j) \in \{(\mathrm{sg}, c), (\mathrm{mot}, r)\}$$

index the two decoder branches, and write

$$S_b = (H_l, H_j).$$

We first consider a rate triple strictly satisfying the five rate inequalities in Theorem 1. Boundary points are obtained by taking limits.

Codebook generation and encoding. Choose a positive rate slack ρ sufficiently small relative to the strict rate margins, and set

$$\widetilde{R}_i = I(H_i; U_i \mid Q) + \rho, \qquad i \in \{c, l, r\}. \tag{22}$$

Fix a deterministic time-sharing sequence q^n whose empirical distribution converges to p_Q.

For each agent i, independently generate

$$N_i = \left\lceil 2^{n\widetilde{R}_i} \right\rceil$$

codewords $u_i^n(\ell_i)$, $\ell_i \in [1 : N_i]$, according to

$$\prod_{t=1}^{n} p_{U_i|Q}(u_{i,t} \mid q_t). \tag{23}$$

The three modality codebooks are independently generated. Assign every index ℓ_i independently and uniformly to one of

$$B_i = \left\lceil 2^{nR_i} \right\rceil$$

bins, and denote the bin assignment by $b_i(\ell_i)$.

After observing h_i^n, encoder i selects the smallest index L_i such that

$$(q^n, h_i^n, u_i^n(L_i)) \in \mathcal{T}_{\epsilon_0}^{(n)}(Q, H_i, U_i). \tag{24}$$

If no such index exists, it selects index 1 and declares an encoding error. The transmitted message is

$$M_i = b_i(L_i). \tag{25}$$

In particular, the selected indices do not depend on the random bin assignments.

For a sufficiently small local typicality tolerance ϵ_0, the covering lemma and (22) imply that the probability of an encoding error tends to zero.

For a fixed typical source sequence, conditioned on successful selection, the selected codeword has the distribution in (23) restricted and normalized to the local conditional typical set. The independent modality codebooks therefore induce independent selections conditional on the source sequences. Choosing ϵ_0 sufficiently smaller than a final joint-typicality tolerance ϵ, the refined Markov lemma implies that

$$\begin{aligned}(q^n, H_c^n, H_l^n, H_r^n, Y_{\mathrm{sg}}^n, Y_{\mathrm{mot}}^n,\\ u_c^n(L_c), u_l^n(L_l), u_r^n(L_r))\end{aligned} \tag{26}$$

is jointly ϵ-typical with probability tending to one.

Decoding and rate constraints. Decoder b receives (M_l, M_j) and searches for a unique index pair $(\widehat{L}_l^{(b)}, \widehat{L}_j)$ in the corresponding bins such that

$$(q^n, u_l^n(\widehat{L}_l^{(b)}), u_j^n(\widehat{L}_j)) \in \mathcal{T}_\epsilon^{(n)}(Q, U_l, U_j). \tag{27}$$

An arbitrary output is used if the search fails or is not unique.

The random-binning packing analysis gives the sufficient conditions

$$\begin{aligned}\widetilde{R}_l - R_l &< I(U_l; U_j \mid Q) - \delta(\epsilon),\\ \widetilde{R}_j - R_j &< I(U_l; U_j \mid Q) - \delta(\epsilon),\\ \widetilde{R}_l + \widetilde{R}_j - R_l - R_j &< I(U_l; U_j \mid Q) - \delta(\epsilon),\end{aligned} \tag{28}$$

where $\delta(\epsilon) \to 0$ as $\epsilon \to 0$. Integer-rounding effects vanish after normalization by n.

To justify the binning step, condition on the sources and codebooks. Since the selected indices are independent of the bin assignments, a competing single-index candidate has bin-collision probability B_i^{-1}. A competing two-index candidate has collision probability $(B_l B_j)^{-1}$. Averaging over the independent codebooks and applying the joint-typicality bounds yields (28).

The product test-channel factorization implies

$$\begin{aligned}I(H_l; U_l \mid Q) - I(U_l; U_j \mid Q)\\ = I(H_l; U_l \mid U_j, Q),\end{aligned} \tag{29}$$

and the analogous identity with l and j exchanged. It also implies

$$\begin{aligned}I(H_l; U_l \mid Q) + I(H_j; U_j \mid Q)\\ - I(U_l; U_j \mid Q)\\ = I(H_l, H_j; U_l, U_j \mid Q).\end{aligned} \tag{30}$$

Consequently, for sufficiently small ρ and typicality tolerances, the strict rate inequalities

$$\begin{aligned}R_l &> I(H_l; U_l \mid U_j, Q),\\ R_j &> I(H_j; U_j \mid U_l, Q),\\ R_l + R_j &> I(H_l, H_j; U_l, U_j \mid Q)\end{aligned} \tag{31}$$

ensure reliable decoding at branch b.

Both branches use the same LiDAR codebook, selected index L_l, and transmitted message M_l. Applying (31) to both branches produces exactly the five rate inequalities of Theorem 1, including the maximum in the LiDAR-rate constraint.

Averaging the sum of the joint-typicality failure probability and the two decoding-error probabilities shows that there exists one deterministic realization of all codebooks and bin assignments for which all three probabilities tend to zero. Fix such a realization. All subsequent entropies and mutual informations refer to the distribution induced by this fixed code.

Distortion. On successful decoding, decoder b forms

$$\widehat{Y}_{b,t} = \varphi_b\big(u_{l,t}(\widehat{L}_l^{(b)}), u_{j,t}(\widehat{L}_j), q_t\big). \tag{32}$$

On the joint-typicality and correct-decoding event, the typical-average lemma bounds the average distortion by its single-letter expectation plus a quantity that vanishes with ϵ. On the complementary event, the distortion is bounded by the maximum of d_b. Therefore,

$$\Delta_b^{(n)} \le \mathbb{E}[d_b(Y_b, \varphi_b(U_l, U_j, Q))] + \xi_{n,\epsilon}, \tag{33}$$

where the nonnegative error term can be chosen so that

$$\lim_{\epsilon \downarrow 0} \limsup_{n\to\infty} \xi_{n,\epsilon} = 0.$$

Below, $\xi_{n,\epsilon}$ may be enlarged to absorb the finitely many typicality, decoding, type-approximation, and rounding terms.

Task relevance. Since decoder b recovers (L_l, L_j) with vanishing error probability, Fano's inequality gives

$$H(L_l, L_j \mid M_l, M_j) \le n\xi_{n,\epsilon}. \tag{34}$$

Moreover, joint typicality and the conditional typical-set cardinality bound imply

$$H(Y_b^n \mid L_l, L_j) \le nH(Y_b \mid U_l, U_j, Q) + n\xi_{n,\epsilon}. \tag{35}$$

The contribution from atypical outcomes is absorbed into $\xi_{n,\epsilon}$, using the finite task alphabet.

Because (M_l, M_j) is a deterministic function of (L_l, L_j),

$$\begin{aligned} H(Y_b^n \mid M_l, M_j) &\le H(L_l, L_j \mid M_l, M_j) \\ &\quad + H(Y_b^n \mid L_l, L_j). \end{aligned} \tag{36}$$

Using $H(Y_b^n) = nH(Y_b)$ and the independence of Q from the source and task variables, we obtain

$$\begin{aligned} \frac{1}{n} I(Y_b^n; M_l, M_j) &\ge H(Y_b) - H(Y_b \mid U_l, U_j, Q) \\ &\quad - 2\xi_{n,\epsilon} \\ &= I(Y_b; U_l, U_j \mid Q) - 2\xi_{n,\epsilon} \\ &\ge \eta_b - 2\xi_{n,\epsilon}. \end{aligned} \tag{37}$$

This argument applies to both tasks.

Operational message redundancy. The index alphabet sizes give

$$H(L_i) \le n\widetilde{R}_i + o(n). \tag{38}$$

For each branch, joint typicality also gives

$$H(S_b^n \mid L_l, L_j) \le nH(S_b \mid U_l, U_j, Q) + n\xi_{n,\epsilon}. \tag{39}$$

Since S_b^n is memoryless and Q is independent of S_b,

$$\begin{aligned} H(L_l, L_j) &\ge I(S_b^n; L_l, L_j) \\ &\ge nI(S_b; U_l, U_j \mid Q) - n\xi_{n,\epsilon}. \end{aligned} \tag{40}$$

By data processing, followed by (38) and (40),

$$\begin{aligned} \frac{1}{n} I(M_l; M_j) &\le \frac{1}{n} I(L_l; L_j) \\ &\le \widetilde{R}_l + \widetilde{R}_j - I(S_b; U_l, U_j \mid Q) \\ &\quad + \xi_{n,\epsilon} \\ &= I(U_l; U_j \mid Q) + 2\rho + \xi_{n,\epsilon}. \end{aligned} \tag{41}$$

The last equality follows from (22) and (30); rounding terms have been absorbed into $\xi_{n,\epsilon}$.

Summing (41) for $j = c$ and $j = r$ yields

$$\begin{aligned} G_{\mathrm{br}}^{(n)} &\le I(U_l; U_c \mid Q) + I(U_l; U_r \mid Q) \\ &\quad + 4\rho + 2\xi_{n,\epsilon} \\ &\le \Gamma + 4\rho + 2\xi_{n,\epsilon}. \end{aligned} \tag{42}$$

Thus the auxiliary-description redundancy condition is sufficient to control the operational message redundancy of the constructed codes.

For any prescribed operational tolerance, choose ρ sufficiently small, then choose the typicality tolerances sufficiently small, and finally choose n sufficiently large. The rate, distortion, relevance, and message-redundancy requirements in the operational definition are then satisfied.

Letting the positive rate margins tend to zero, taking the union over all admissible finite auxiliary distributions and reconstruction functions, and taking its closure proves

$$\mathcal{R}_{\mathrm{RDRR}}^{\mathrm{in}}(\mathbf{D}, \boldsymbol{\eta}, \Gamma) \subseteq \mathcal{R}_{\mathrm{RDRR}}^{*}(\mathbf{D}, \boldsymbol{\eta}, \Gamma).$$

□

Appendix B
Proof of the Outer Bound

Proof. Consider an achievable operational tuple

$$(\mathbf{R}, \mathbf{D}, \boldsymbol{\eta}, \Gamma).$$

By the definition of achievability, there exists a sequence of deterministic distributed codes with vanishing tolerances ε_n such that

$$\begin{aligned} \frac{1}{n} \log_2 |\mathcal{M}_i| &\le R_i + \varepsilon_n, \quad i \in \{c, l, r\}, \\ \Delta_b^{(n)} &\le D_b + \varepsilon_n, \quad b \in \{\mathrm{sg}, \mathrm{mot}\}, \\ \frac{1}{n} I(Y_b^n; M_l, M_j) &\ge \eta_b - \varepsilon_n, \quad (b, j) \in \{(\mathrm{sg}, c), (\mathrm{mot}, r)\}, \\ G_{\mathrm{br}}^{(n)} &\le \Gamma + \varepsilon_n. \end{aligned} \tag{43}$$

The final constraint is part of the operational problem, but will not be used to restrict the outer auxiliary region.

For a generic branch

$$(b, j) \in \{(\mathrm{sg}, c), (\mathrm{mot}, r)\},$$

define

$$S_{b,t} = (H_{l,t}, H_{j,t}), \qquad S_b^{t-1} = (H_l^{t-1}, H_j^{t-1}), \tag{44}$$

and

$$V_{l,t}^{(b)} = (M_l, S_b^{t-1}), \qquad V_{j,t} = (M_j, S_b^{t-1}). \tag{45}$$

Let Q be uniform on $\{1, \ldots, n\}$ and independent of all source sequences and messages. Set

$$\begin{aligned} H_i &= H_{i,Q}, & Y_b &= Y_{b,Q}, \\ U_l^{(b)} &= (V_{l,Q}^{(b)}, Q), & U_j &= (V_{j,Q}, Q). \end{aligned} \tag{46}$$

The same time index Q is used for both branches. These definitions give all four auxiliaries on one common probability space with the prescribed source and task marginal distribution.

Auxiliary-variable conditions. Fix $Q = t$. All source and task samples outside time t are independent of the full current source and task tuple. Since the encoders are local, (M_l, S_b^{t-1}) depends on the current tuple only through $H_{l,t}$, and (M_j, S_b^{t-1}) depends on it only through $H_{j,t}$. Their pair depends on the current tuple only through $(H_{l,t}, H_{j,t})$.

Because Q is independent and the source law is identical across time, adjoining Q preserves these conditional independence statements. Consequently,

$$\begin{aligned} U_l^{(b)} &- H_l - (H_j, Y_b), \\ U_j &- H_j - (H_l, Y_b), \\ (U_l^{(b)}, U_j) &- (H_l, H_j) - Y_b. \end{aligned} \tag{47}$$

The shared source history and time index generally prevent a product factorization of the two auxiliaries conditioned on their current sources. No such factorization is needed for the outer bound.

Individual rates. Since M_l is a deterministic function of H_l^n, and hence of S_b^n,

$$\begin{aligned} n(R_l + \varepsilon_n) &\geq H(M_l \mid M_j) \\ &= I(S_b^n; M_l \mid M_j) \\ &= \sum_{t=1}^{n} I(S_{b,t}; M_l \mid M_j, S_b^{t-1}) \\ &= \sum_{t=1}^{n} I(S_{b,t}; V_{l,t}^{(b)} \mid V_{j,t}) \\ &= nI(H_l, H_j; U_l^{(b)} \mid U_j). \end{aligned} \tag{48}$$

The last equality follows from (46); both auxiliaries contain Q.

Interchanging the roles of the two messages gives

$$R_j + \varepsilon_n \geq I(H_l, H_j; U_j \mid U_l^{(b)}). \tag{49}$$

Pairwise sum rates. The message pair (M_l, M_j) is a deterministic function of S_b^n. Thus

$$\begin{aligned} n(R_l + R_j + 2\varepsilon_n) &\geq H(M_l, M_j) \\ &= I(S_b^n; M_l, M_j) \\ &= \sum_{t=1}^{n} I(S_{b,t}; M_l, M_j \mid S_b^{t-1}) \\ &= \sum_{t=1}^{n} I(S_{b,t}; M_l, M_j, S_b^{t-1}) \\ &= nI(H_l, H_j; U_l^{(b)}, U_j). \end{aligned} \tag{50}$$

The penultimate equality uses the independence of $S_{b,t}$ and S_b^{t-1}. The final equality also uses the independence of $(H_{l,Q}, H_{j,Q})$ and Q.

Applying (48)–(50) to both branches gives all five rate constraints in (18), up to vanishing tolerances. In particular, the same message M_l must satisfy both LiDAR lower bounds, yielding the maximum in the LiDAR-rate constraint.

Reconstruction and distortion. To use a common ordering of the message pair, write

$$\overline{\psi}_{\mathrm{sg}}^{(n)}(m_l, m_c) = \psi_{\mathrm{sg}}^{(n)}(m_c, m_l), \qquad \overline{\psi}_{\mathrm{mot}}^{(n)}(m_l, m_r) = \psi_{\mathrm{mot}}^{(n)}(m_l, m_r).$$

The pair $(U_l^{(b)}, U_j)$ contains (M_l, M_j, Q). It therefore permits the deterministic reconstruction

$$\begin{aligned} \widehat{Y}_b &= \varphi_b^{\mathrm{out}}(U_l^{(b)}, U_j) \\ &= \big[\overline{\psi}_b^{(n)}(M_l, M_j)\big]_Q. \end{aligned} \tag{51}$$

Consequently,

$$\begin{aligned} \mathbb{E}[d_b(Y_b, \widehat{Y}_b)] &= \frac{1}{n}\sum_{t=1}^{n} \mathbb{E}[d_b(Y_{b,t}, \widehat{Y}_{b,t})] \\ &= \Delta_b^{(n)} \\ &\leq D_b + \varepsilon_n. \end{aligned} \tag{52}$$

Task relevance. Let $B_b = (M_l, M_j)$. Since the source is memoryless and B_b is a deterministic function of S_b^n,

$$Y_b^{t-1} - S_b^{t-1} - (B_b, Y_{b,t}) \tag{53}$$

is a Markov chain. In particular,

$$I(Y_{b,t}; Y_b^{t-1} \mid B_b, S_b^{t-1}) = 0.$$

Using the temporal independence of the task sequence, we obtain

$$\begin{aligned} I(Y_b^n; B_b) &= \sum_{t=1}^{n} I(Y_{b,t}; B_b \mid Y_b^{t-1}) \\ &= \sum_{t=1}^{n} I(Y_{b,t}; B_b, Y_b^{t-1}) \\ &\leq \sum_{t=1}^{n} I(Y_{b,t}; B_b, Y_b^{t-1}, S_b^{t-1}) \\ &= \sum_{t=1}^{n} I(Y_{b,t}; B_b, S_b^{t-1}) \\ &= nI(Y_b; U_l^{(b)}, U_j). \end{aligned} \tag{54}$$

The fourth line follows from (53). The final equality uses (46) and the independence of $Y_{b,Q}$ and Q.

Combining (54) with the operational relevance requirement gives

$$I(Y_b; U_l^{(b)}, U_j) \geq \eta_b - \varepsilon_n. \tag{55}$$

Thus both relevance constraints admit the single-letter form used in Theorem 2.

Redundancy relaxation and closure. The operational code also satisfies the prescribed message-redundancy budget. The outer region relaxes this constraint by omitting it. In particular, no inequality of the form

$$I(U_l^{(\mathrm{sg})}; U_c) + I(U_l^{(\mathrm{mot})}; U_r) \leq \Gamma$$

is inferred from the message-redundancy constraint: the converse auxiliaries contain shared source histories and the time index.

For each code in the sequence, the constructed auxiliaries have finite alphabets and satisfy the Markov conditions exactly. Equations (48), (49), (50), (52), and (55) show that the tuple with coordinates

$$R_i + \varepsilon_n, \qquad D_b + \varepsilon_n, \qquad \max\{\eta_b - \varepsilon_n, 0\}$$

belongs to $\mathscr{O}$. These coordinates converge to $(\mathbf{R}, \mathbf{D}, \boldsymbol{\eta})$. Taking closure in the full performance space therefore yields

$$(\mathbf{R}, \mathbf{D}, \boldsymbol{\eta}) \in \overline{\mathscr{O}},$$

and hence

$$\mathcal{R}^{*}_{\mathrm{RDRR}}(\mathbf{D}, \boldsymbol{\eta}, \Gamma) \subseteq \mathcal{R}^{\mathrm{out}}_{\mathrm{RDRR}}(\mathbf{D}, \boldsymbol{\eta}).$$

Together with Appendix A, this establishes (20). Since $\mathcal{R}^{\mathrm{out}}_{\mathrm{RDRR}}(\mathbf{D}, \boldsymbol{\eta})$ is closed for fixed $(\mathbf{D}, \boldsymbol{\eta})$, the inclusion also holds for the closure of the achievable rate triples. □